

PAPER

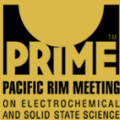
Low temperature dry etching of chromium towards control at sub-5 nm dimensions

To cite this article: Daniel Staaks *et al* 2016 *Nanotechnology* **27** 415302

View the [article online](#) for updates and enhancements.

You may also like

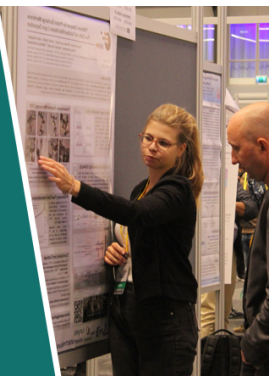
- [Maximum likelihood versus likelihood-free quantum system identification in the atom maser](#)
Catalin Catana, Theodore Kypraios and Mdlin Gu
- [Reconfiguring quantum error-correcting codes for real-life errors](#)
Muhammad Ahsan and Syed Abbas Zilqurnain Naqvi
- [Analytical eigenstates for the quantum Rabi model](#)
Honghua Zhong, Qiongtao Xie, Murray T Batchelor et al.



HONOLULU, HI
Oct 6–11, 2024

Abstract submission deadline:
April 12, 2024

Learn more and submit!



Joint Meeting of

The Electrochemical Society
•
The Electrochemical Society of Japan
•
Korea Electrochemical Society

Low temperature dry etching of chromium towards control at sub-5 nm dimensions

Daniel Staaks^{1,2}, XiaoMin Yang³, Kim Y Lee³, Scott D Dhuey¹,
Simone Sassolini¹, Ivo W Rangelow² and Deirdre L Olynick¹

¹ Molecular Foundry, Lawrence Berkeley National Laboratory, Berkeley, 94720, USA

² Ilmenau University of Technology, Dept. of Micro- and Nanoelectronic Systems, D-98684, Germany

³ Fremont Research Center, Seagate Technology, Fremont, CA 94538, USA

E-mail: dlolynick@lbl.gov

Received 13 April 2016, revised 21 July 2016

Accepted for publication 18 August 2016

Published 8 September 2016



CrossMark

Abstract

Patterned chromium and its compounds are crucial materials for nanoscale patterning and chromium based devices. Here we investigate how temperature can be used to control chromium etching using chlorine/oxygen gas mixtures. Oxygen/chlorine ratios between 0% and 100% and temperatures between -100°C and $+40^{\circ}\text{C}$ are studied. Spectroscopic ellipsometry is used to precisely measure rates, chlorination, and the thickness dependence of n and k . Working in the extremes of oxygen content (very high or very low) and lower temperatures, we find rates can be controlled to nanometers per minute. Activation energies are measured and show that etch mechanisms are both temperature and oxygen level dependent. Furthermore, we find that etching temperature can manipulate the surface chemistry. One surprising consequence is that at low oxygen levels, Etching rates increase with decreasing temperature. Preliminary feature-profile studies show the extremes of temperature and oxygen provide advantages over commonly used room temperature processing conditions. One example is with higher ion energies at -100°C , where etching products deposit.

 Online supplementary data available from stacks.iop.org/NANO/27/415302/mmedia

Keywords: chromium, inductively coupled plasma, dry etching, low temperature, bit patterned media, ICP, oxygen concentration

(Some figures may appear in colour only in the online journal)

1. Introduction

Chromium and its oxides have been important in fabrication and nanotechnology for many decades. While patterning lithographic photomasks have been the most prominent application, many more applications have emerged. For instance, chromium is an important etching mask for transferring lithographic patterns to silicon and oxides [1, 2]. In addition, patterned chromium has been used as DUV laser direct writing masks [3] and as a seed layer for guided self-assembly of high density magnetic media [4]. In the year prior to this publication, gate-all-around transistors on semiconductor nanowires were fabricated using patterned Cr gates [5]. Moreover, patterned CrO_2 has been recognized as a potential spintronics material [6, 7]. Here, we are targeting

chromium as enabling technology for the high throughput production of magnetic bit-patterned media (BPM) at densities greater than 5 TB/in^2 .

Bit patterned media in combination with heat-assisted magnetic recording (HAMR) is slated to bring hard drive densities to the 7 TB/in^2 density range. Isolated bits will have dimensions below 6 nm and will be patterned using nanoimprint technology [4, 8, 9]. Template fabrication at this resolution is complex and culminates in the etching of quartz and fused silica with a hard mask, chromium being a prime hard mask candidate. While plasma etching of chromium and its oxides is now almost 40 years old, the range of conditions explored has been limited. The focus application has been optical lithography projection masks [1, 2, 10]. However, the critical dimension/defect requirements of optical masks are

far less demanding than in nanoimprint masks—nanoimprint lithography requires chromium features patterned at 1x the pattern size, while optical projection masks print with a 4x reduction with mask dimensions near 100 nm. Abe *et al* [11] first reported that chromium can be etched by chlorine containing plasmas. Later, oxygen was found to play an important role [12, 13]. Predominantly, work has focused on this chemistry, although chromium can also be etched with other halogens such as fluorine. Difficulties found in the chromium etching process were overcome in large part by changing the dimensions of the original resist mask pattern to compensate for mask erosion and undercut. This approach is not feasible for sub-10 nm patterning of nanoimprint templates. In order to pattern with such tight controls, new approaches to chromium etching must be explored.

A strong knob for controlling etch rates and profile is temperature [14]. Improvements of the cryogenic process on profile control towards high aspect ratio etching and critical tasks thereof [15] have been extensively studied for silicon [16–19] but are not studied in great detail for other materials.

In this work, we investigate low-temperatures as a means to gain unprecedented control of the chromium etching process. In addition, we explore the role oxygen plays in chlorine/oxygen etching chemistries over the range of 0%–100% oxygen. Although oxygen partial pressure is known to be important, previously only low oxygen partial pressures were studied. The maximum studied was approximately 25% of the total Cl_2/O_2 flow, because these levels give better selectivity to the carbon based resist masks. Nanoimprint mask fabrication is not limited to carbon based masks, and so we study a large range of oxygen partial pressures.

In this paper, ICP etching of chromium is studied at substrate temperatures between -100°C and $+40^\circ\text{C}$ over a large range of oxygen/chlorine ratios; the superiority of ICP over capacitively controlled reactive ion etching has been mentioned by several authors [20–22]. Etching rates are characterized using spectroscopic ellipsometry (SE). SE is not trivial to implement for thin absorbing films such as chromium where n and k change as a function of thickness. However, by studying thin chromium films on carefully chosen optical stacks, we use SE as a fast, non-invasive technique to measure film thickness. This allows the collection of significant amounts of data.

Overall, we find that we can use low temperature to manipulate the etching rates and surface chemistry for benefits such as increased selectivity and profile control. Moreover, working in the extremes of temperature ($\sim -50^\circ\text{C}$) and oxygen (high or low partial pressures), we find better rate/process control with sub-nanometer accuracy. However, mechanisms vary significantly as a function of oxygen and temperature. The role of sticking coefficients and competing surface reactions are discussed. Furthermore, we find that chromyl chlorides condense at low temperature and can be used to control profiles, although the etching rates are extremely low.

2. Experimental

Samples: Blanket films—Chromium 20 nm (± 1.5 nm) was electron beam evaporated⁴ onto 250 nm (± 12.5 nm) dry thermal oxide on a 4 inch Single Side polished prime silicon wafer. Wafers were cleaved into chips of $1 \times \frac{1}{2}$ inches in size and thickness measured (SE) in the same locale pre and post-etching. Chips were placed in the center of a silicon carry wafer during the etching process. For improved thermal contact to the carrier wafer, a perfluoropolyether vacuum oil (Fomblin25/6) was used to enhance chip to sample thermal contact. Fomblin was carefully applied under the sample but not exposed to the plasma.

Patterned films—For preliminary results on patterned films, we used e-beam lithography (Vistec VB300). Hydrogen silsesquioxane (HSQ) resist was spun on a 30 nm chromium layer on 250 nm thermal oxide (4" Wafer). Post exposure the resist was developed using NaOH based developer. No hard bake was performed to prevent shrinkage of the resist. Subsequently the HSQ masked chromium was etched and cross-section imaged by scanning electron microscopy (SEM).

Etching: Samples were etched in an inductively coupled plasma (ICP) reactor (Oxford Plasmalab 150 Cobra). The wafer is mechanically clamped onto a temperature-controlled biased electrode. Helium backside cooling is used for thermal conduction between the wafer and electrode temperatures. Electrode temperature, measured via a thermocouple buried within the electrode, is controlled to $\pm 1^\circ\text{C}$.

The etching recipe consists of three steps: wafer cooling, strike and main etch. The main etch was conducted at 5 W table power and 500 W ICP power at 5 mTorr chamber pressure using O_2/Cl_2 gas mixture with a total flow of 40 sccm. To ignite the plasma, a strike step was conducted with 30 W table power for 5 s. Prior to the strike, the wafer is brought to temperature with a cooling step. At these low plasma powers, plasma induced temperature increase of the wafer is negligible. After removing from the etcher the sample thickness was measured via SE. Total time from removal of the sample, to etcher to SE was less than 5 min to reduce oxidation-effects. Chromium samples were etched to leave between 6–10 nm films on the surface. Etch rate per minute was determined by calculating $\left[\frac{(d_{\text{prior}} - d_{\text{after}})}{t} \right] * 60$ (t in seconds), where d_{prior} is the thickness of the chromium before etching and d_{after} post-etching. Measurements were made within 1 mm of the same locale on the chip.

Reproducibility was tested by etching multiple samples at same condition (0°C 50% oxygen). The final film thicknesses had a standard deviation of 0.5 nm. As higher errors can be expected at shorter etch times, this can be assumed to be the order of error we can expect leading to etch rate uncertainties of 3%–10% depending on temperature and oxygen concentration.

⁴ Cr actually sublimates, but as it is mostly referred to as evaporator, we use this term.

Note, etching in chlorine based chemistry is very sensitive to the chamber wall condition [23–25]. Chamber pre conditioning with the etch gas composition was performed for 5 min on a silicon wafer, prior to etching actual samples.

Thin film measurement via SE: To explore the chromium etching mechanism over a large range of temperatures and oxygen content, we needed a fast and accurate means to measure etching rates. We detail our method of film measurement here to reinforce the validity of our results. We use phase modulated SE (Horiba UVISEL) which is not trivial to implement for thin, absorbing films [26]. Here we describe the issues and how we overcame them to study chromium etching rates using SE.

2.1. SE: challenges with absorbing films

Ellipsometry measures changes in light polarization after reflection from a surface. The change in polarization between the incident and reflected light at angle of incidence, θ is characterized by two values, the phase difference Δ and the amplitude ratio component Ψ . These values are connected by the well known equation [27] of the complex reflectance ratio, ρ , of p- and s- polarized light, where

$$\rho = \frac{r_p}{r_s} = \left(\frac{E_{rp}}{E_{ip}} \right) / \left(\frac{E_{rs}}{E_{is}} \right) = \tan(\Psi) \exp(i\Delta). \quad (1)$$

Measurements of film refractive index (n), extinction coefficient (k) and thickness (t) via ellipsometry are indirect in that a model of the film structure is necessary to determine these film properties from the Ψ , Δ measurements. For a single wavelength measurement, only two unknowns can be determined. With SE, a new Ψ , Δ is measured at each wavelength within a measured spectrum. If the film is transparent, that is $n = f(\lambda)$ and $k = 0$, wavelength is independent of thickness. Under this transparency condition, n and t can be fit for multiple layers in a stack even if the layers are different compositionally (e.g. oxide 1 and oxide 2) and/or structurally (oxide 1 split into layers with and without roughness/surface oxide).

For absorbing films, the problem is more difficult. If the film is too thick, light will not penetrate and no measurement is made. Even for thin absorbing films, where light can penetrate the films, there is a problem determining n and k . In this case, the optical constants are thickness dependent, i.e. $n = f(\lambda, t)$ and $k = f(\lambda, t)$. As a consequence accurate determination of n , k and t are difficult. Fortunately, accurate chromium thin film measurements can be made by enhancing the model sensitivity as described in the subsequent section.

2.2. Chromium optical model generation and verification

As noted by Hilfiker *et al* [26], several techniques can be used to improve the accuracy and sensitivity of the SE model for chromium thin film measurements. Their paper focuses on sensitivity enhancement by measurement at multiple angles. Here we show that interference enhancement, optical constant parametrization, and spectral range limitation creates an accurate and sensitive model. This method decorrelates

optical constants and thickness such that multiple angle measurements are not necessary, speeding the measurement process. In addition, we evaluated the spectral range and thicknesses for highest transmission (see supplement A) and found that we could work with films up to 20 nm.

How we came to the appropriate parameters is described in the next section. Overall, we find we can make fast, accurate SE measurements of etched sub-20 nm chromium samples on 250 nm silicon oxides over 300–410 nm wavelengths. Hence all etching measurements with SE are done with sub 20 nm films on 250 nm of oxide.

2.3. Interference enhancement and spectral range for the measurement

Interference enhancement produces spectral features in the ellipsometric measurements that improve the chromium film measurement sensitivity. Chromium films were evaporated on 250 nm (± 12.5 nm) dry thermal oxide to create this enhancement. Figure 1(a) shows Δ and Ψ measurements for chromium/oxide stacks with varying oxide thicknesses. Interference in the oxide film leads to 1 minimum and 1 maximum. With increasing chromium film thickness (from 0 to 25 nm) we see the interference oscillations are damped and the minimum is shifted. This damping and shift improves sensitivity of the SE model, particularly in the sub 420 nm regime where the oscillations are apparent. The benefit of this improvement is shown in figures 1(b) and (c) where Δ , Ψ points are plotted. Each curve represents a different thickness at a constant wavelength. For the smaller wavelengths (figure 1(b)), where the interference fringes are present, the points are much better separated allowing one to more easily converge to a unique solution. Crowded areas in the smaller wavelengths plot represent the thicker films where the interference fringes are damped. As an example the thickness for 350 nm wavelength is noted. For the larger wavelengths, the points are less separated (figure 1(c)) for all thicknesses. This makes it difficult to converge on a unique thickness solution, if one exists. Hence, for all our SE measurements, we fit the data in the sub 410 nm wavelength regime, which correlates to a phase difference of 2π .

Typically the Brewster angle is chosen for ellipsometric measurement, because the largest changes in polarization can be measured as the p component goes to zero. However, the Pseudo Brewster angle of absorbing film stacks is wavelength dependent [28]. We find the most model sensitivity is obtained between 300 and 400 nm range and at 70° , close to the pseudo Brewster angle (see also supplement B).

2.4. Optical model and layer parameterization

To reduce the number of fit parameters in the final stack measurements, we use optical constant parametrization. The layer structure was assumed to be as follows:

Layer 3: Surface layer consisting of chromium oxide and surface roughness

Layer 2: Chromium film

Layer 1: Silicon Oxide

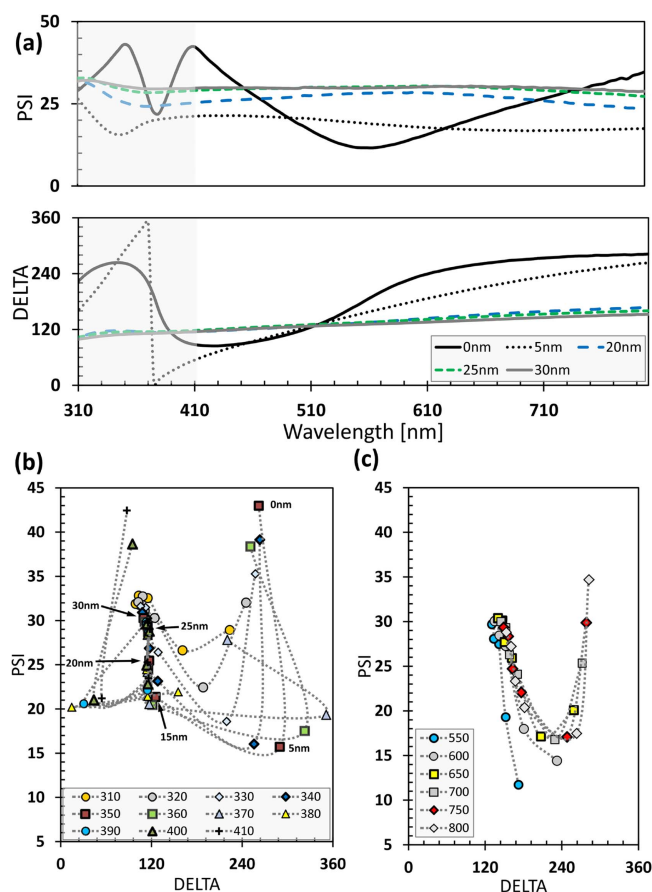


Figure 1. Maximizing spectroscopic ellipsometry (SE) sensitivity for accurate measurement of chromium film etching rates by measuring in the 310–410 nm range. Chromium (0–30 nm) is deposited on 250 nm SiO₂/Si wafer substrates. (a): Psi and Delta as a function of wavelength and chromium film thickness. Interference enhancement improves the signal particularly in the wavelength range of 310–410 nm (shaded area). This is shown more clearly in graphs (b) and (c) which plots Psi as a function of Delta for Cr sample thickness between 0 and 30 nm in 5 nm increments. Curves show values for one wavelength as thickness is varied. Ideally, points should be well separated to find a unique, accurate solution. (Curves are smoothed to guide the eye, but do not represent an interpolation.) This case is satisfied for the wavelength range in (b), 310–410 nm, but not (c) which has a wavelength range of 550–800 nm.

Substrate: Silicon

The model for the layer stack was created by using an iterative approach to reduce the fit parameters of the final structure. First the silicon substrate was measured without oxide on the top. These measurements were then input into the silicon substrate model. Next, the oxide layer was measured and extracted values input into the silicon oxide model. The chromium layer was modeled using a classical dispersion equation, consisting of a Lorenz oscillator and a Drude term.

To test the model, different thicknesses of evaporated chromium were used. The chromium was first modeled as a single layer without oxide on the top and without roughness to get a first estimation of the parameters and then the surface layer was added. The surface layer was assumed to be chromium oxide and was fit with a Tauc–Lorenz model. The best fit was achieved by modelling the layer as chromium oxide with surface roughness. Using the measured starting

parameters, all the layers were fit simultaneously. The output was used as a starting point for the next fit. Iterations were conducted until there was no further change in the fit. Finally, n , k and t were extracted as a function of thickness.

To verify the surface layer model held for etched films, a set of chromium films were etched to different thicknesses and the same model applied. As refractive index n and extinction coefficient k of metals are a function of thickness [29], a comparison of calculated n and k between as deposited chromium and post etched chromium confirmed the same curve within error bars (supplement F). Thus we assume the surface layer is the same in both the evaporated and etched samples.

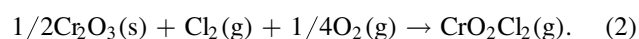
3. Chromium etching behavior with temperature and oxygen

Blanket film etching

Figure 2 shows etching rates as a function of oxygen partial pressure and temperature between +40 °C and –80 °C. Lower temperatures were also explored, down to –100 °C, below the freezing point of chromyl chloride and are discussed in section 4. With changes in oxygen content, the etching rates are highest in the 20%–30% oxygen regime in agreement with Kang *et al* [30]. The rates decrease towards the extremes of high or low oxygen (concave shape of the curves). At the extremes of oxygen level, 0% or 100% the etching rates are very low. At 20 °C these were comparable to the sputter yield with 40 sccm Argon plasma where the rate is far below 1 nm min^{–1}. With just a small amount of oxygen in the Cl₂/O₂ mixture, the etching increases dramatically as observed by other authors [31].

Generally, etching rates decrease with decreasing temperature, except in the low oxygen regimes where we see a crossover. When oxygen content is low, low temperature promotes higher etching rates than high temperature. Ultimately we can see that etching rates can be reduced to the nanometer scale for controlled etching of very thin films.

To understand the peak etching rate position (between 25% and 50% oxygen), we look at the proposed etching mechanism for chromium. It has been shown by Gaballah *et al* [32] that thermally activated reactions of chromium with oxygen and chlorine have a maximum etch rate at oxygen partial pressures of 20%. In a high temperature, non-plasma environment, oxygen is presumed to first oxidize the surface. Subsequently, this oxidized surface is etched by chlorine gas. The proposed reaction is



This leads to a stoichiometric need of 25% oxygen for a pre-oxidized surface and 50% if oxidizing and etching simultaneously.

Nakata *et al* studied chromium etching mechanism in the plasma environment using Auger Electron Spectroscopy. They proposed that oxygen and chlorine radicals (*) interact with the chromium and chromium oxides to form a CrO₂Cl₂

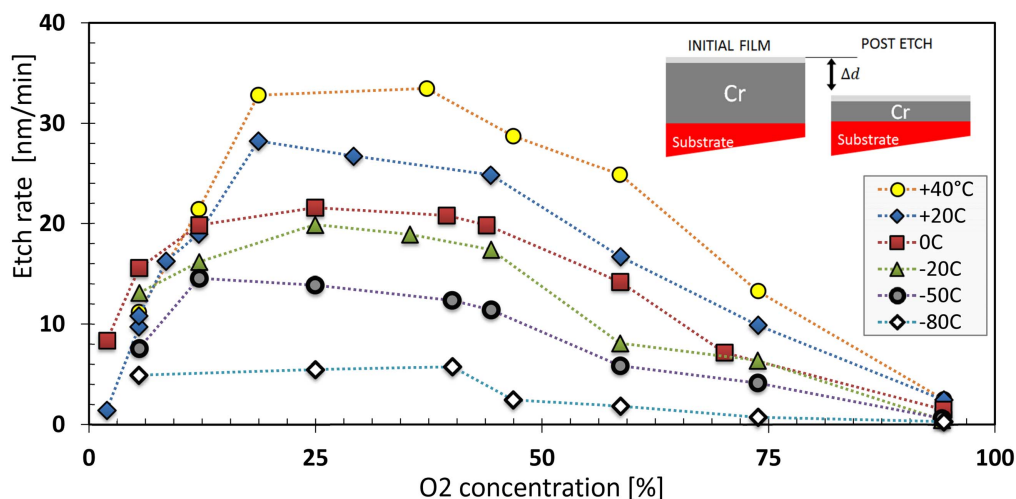


Figure 2. Chromium etch rate in nm min^{-1} as function of temperature and oxygen concentration in the oxygen/chlorine mixture. Each curve represents one temperature. Etch rate was determined as $d/t \cdot 60$, the change in thickness over the etching time. Chromium oxide layers were included in the thickness measurement.

volatile etching product:



and



Later, the specific role of oxygen was investigated by Kwon *et al* [13] who found evidence of CrO_xCl_y compounds as well as Cr–Cl and Cr–O bond formation on the etched Cr surface. Furthermore, they found Cr etching rates increased rapidly with oxygen concentration. Oxygen partial pressure was subsequently found to be a dominant parameter in the chromium etching [22].

Note that Kang *et al* [30] also thought that CrOCl may be a product due to the presence of optical spectroscopy peaks related to Cr^{3+} . However, Schäfer *et al* [33] found that CrOCl is a very stable product in 1×10^{-5} Torr vacuum. It does not sublime and decomposes to Cr_2O_3 and gaseous CrCl_3 at temperatures greater than 800°C . In addition, a comprehensive analysis of chromium species has been done by Ebbinghaus *et al* [34]. The chromyl chloride CrO_2Cl_2 has a high exothermic enthalpy: $\Delta H_f^0(298\text{K}) = -519.2 \pm 4.2 \text{ kJ mol}^{-1}$. CrOCl may be a product or it may form from CrO_2Cl_2 in the gas phase. In any case, it is not clear that this is product of the surface reaction. Hence we focus on CrO_2Cl_2 in the subsequent discussion.

The generation of CrO_2Cl_2 in a plasma environment is described, in equations (2)–(4). The shift of the peak etch rate from 50% in a thermal environment to lower oxygen concentrations in a plasma environment is likely due to the relative concentration of the oxygen and chlorine species on the surface. This concentration will depend in large part on the relative sticking coefficients. The importance of sticking coefficient is reinforced by the increase of etching rate with decreasing temperature in the low oxygen regime. Sticking coefficients are expected to increase with decreasing temperature. The increase in rates at low temperature and low

oxygen is suggestive in that the oxygen sticking coefficient increases more than the chlorine sticking coefficient with decreasing temperature.

As mentioned earlier, etch rates increase very strongly with introduction of oxygen into the chamber. We calculated the total flux of oxygen and chlorine to the surface and used reference data by Hoekstra *et al* [35] to derive the radical flux to the surface. We assumed CrO_2Cl_2 is the only volatile product and then calculated the amount of oxygen needed to give the measured etch rates from figure 2. Resulting calculations can be seen in supplement E and figure E1 and show that sufficient oxygen radicals are available to account for the etching. This supports etching is much more likely according to equations (3) and (4) rather than equation (2).

In terms of reproducibility it turned out that BCl_3 had an effect on etch rates. Measurements after BCl_3 use show slightly higher etch rates (supplement D). Most likely chamber wall recombination was affected. It is known that BCl_3 can affect chamber wall conditioning [36]. Chamber wall recovery took significant time to bring etch rates back close to the previous condition. After using BCl_3 plasma, we see all resulting curves shifted to slightly higher etch rates but with the same trends in overall behavior, for instance, the increase in etching rates with decreasing temperature for the low oxygen concentrations. Etch rates were repeatable once the chamber conditioning was recovered with significant seasoning of the chamber.

Arrhenius studies

Figures 3(a) and (b) show data from figure 2 as Arrhenius plots at different oxygen partial pressure. The data is separated into 2 plots for easier observation. Linearity is clear for the oxygen partial pressures of 25% and 35% to temperatures down to -50°C . In addition, these oxygen levels have slopes (activation energies) that are indistinguishable within our measurement error. In the other curves, linearity is not

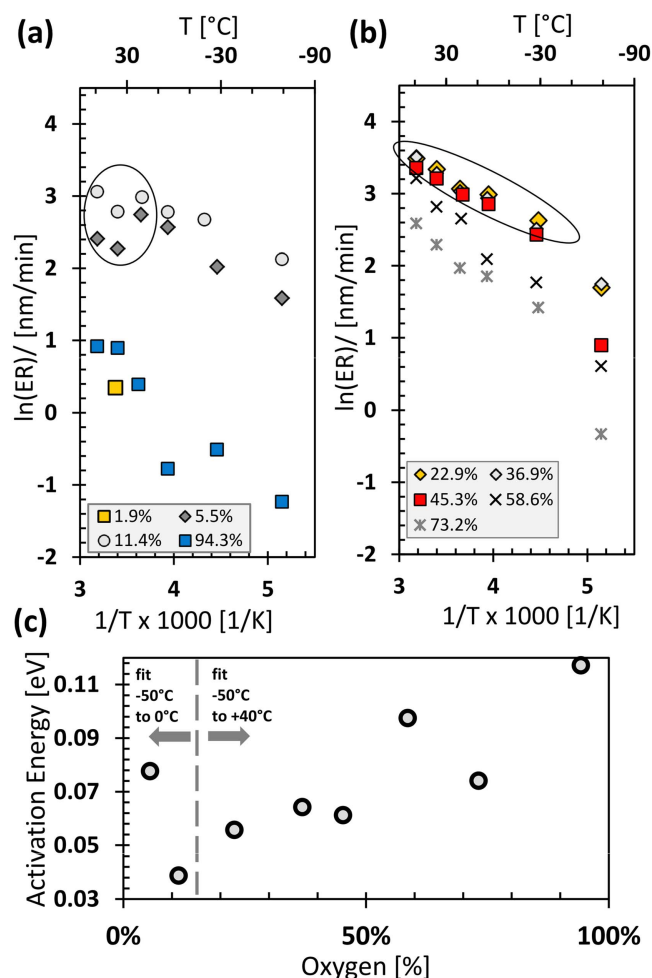


Figure 3. Arrhenius plots of chromium etching data ((a) and (b)) and activation energy as a function of oxygen concentration (c). (a) Low and high oxygen concentrations. Circled data shows etch rates drop at high temperature for 11.4% and 94.3%. (b): mid oxygen concentrations. Concentrations between 22% and 45% show strong linear behavior with the same activation energy. Ellipse shows linear region for E_A fit. (c): activation energy plotted as a function of oxygen concentration. Activation shows increase with oxygen. In the low oxygen concentration, the E_A fit was from -50°C to 0°C . In the high oxygen concentration, the fit was over -50°C to $+40^\circ\text{C}$ temperature range.

apparent over this whole range suggesting that the mechanisms change with both temperature and oxygen concentration.

There are clear discontinuities in the data for the low oxygen concentrations (data points circled in figure 3(a)) which as already mentioned, may be due to changes in the sticking coefficient. In addition, we see discontinuities in the higher oxygen regime (right) where the etching rates fall off faster than expected at -80°C . We confirmed these discontinuities were repeatable (supplement D) and were not related to errors in temperature measurement from the tool imbedded thermocouple.

This apparent change in activation energy (slope) as temperature changes needs further investigation. For now, we look at activation energy as a function of oxygen content for the regions that appear predominantly linear (figure 3(c)). Data plotted for the low oxygen regime was fit only over the

points from -50°C to 0°C . All other data was fit from -50°C to $+40^\circ\text{C}$ to find activation energy. Generally, we see an increase in activation energy with increasing oxygen content. The activation energy increases from 0.04 eV at 12% to 0.12 eV at the highest oxygen concentration in absolute numbers. This is much lower than the activation rates needed in a non-plasma environment. For instance, Tonotani *et al* [37] measured an activation energy of 0.41 eV for chromium oxide in oxygen gas only using temperatures greater than 200°C , while Gaballah *et al* [32] have measured the activation energy between 0.4 and 0.9 eV in the temperature range of 550°C – 1000°C and $\text{Cl}_2/\text{O}_2 = 4$.

Overall, the activation energy in this plasma environment is on the order of physisorption processes on the surface. This activation energy is a superposition of all energies needed or gained between introduction of the active species and the desorption of the volatile product. Generally, radicals are more reactive and thus the activation energy will be lower in a plasma environment. It is known that desorption rates and sticking coefficients can be coverage dependent which would in part explain the activation energy changes with different oxygen partial pressures [38, 39]. In addition, species not shown in the primary reaction (equations (3) and (4)), may be predominant at the surface at the extremes of oxygen concentration, for instance the non-volatile chromium chlorides when the oxygen content is low. These low activation energies suggest that that one or more of the partial reactions involved in the etching process is exothermic. For instance, as mentioned previously, chromium oxidation is an exothermic process.

From our data we can conclude that the etching of chromium with chlorine and oxygen can be separated into three different regions, depending on the oxygen concentration. Most likely the low oxygen region rate is dominated by the need to remove chromium chloride compounds at the surface. In fact, when oxygen is removed and the forward power is reduced to produce a 2 V DC bias, the chromium film grows. This might be due to the incorporation of chlorine into the chromium, leading to the formation of a chromium chloride compound. At higher powers, the film does not grow suggesting the removal of Cl_2 before reaction. In fact, we saw that etching rates can decrease with increasing power under low pressure conditions.

In the high oxygen regime the oxygen most likely forms chromium oxide compounds that interact with chlorine. Hence a chromium oxygen bond break may be needed for the volatile species to form, increasing the activation energy. Our ellipsometry model results show that chromium oxide stabilizes at 2.5 nm thickness at lower power in contrast to 1.2 nm on as deposited samples. This suggests oxygen does diffuse through the surface chromium oxide to grow a thicker film. This is consistent with the self-passivation oxide known in chromium oxide films. In the mid-oxygen regime oxygen and chlorine are sufficiently present at the surface so that there is a higher likelihood of forming the volatile chromyl chloride.

The change of slope with temperature needs to be investigated further, but as mentioned previously, for the low oxygen/high temperature regime, may be related to changes

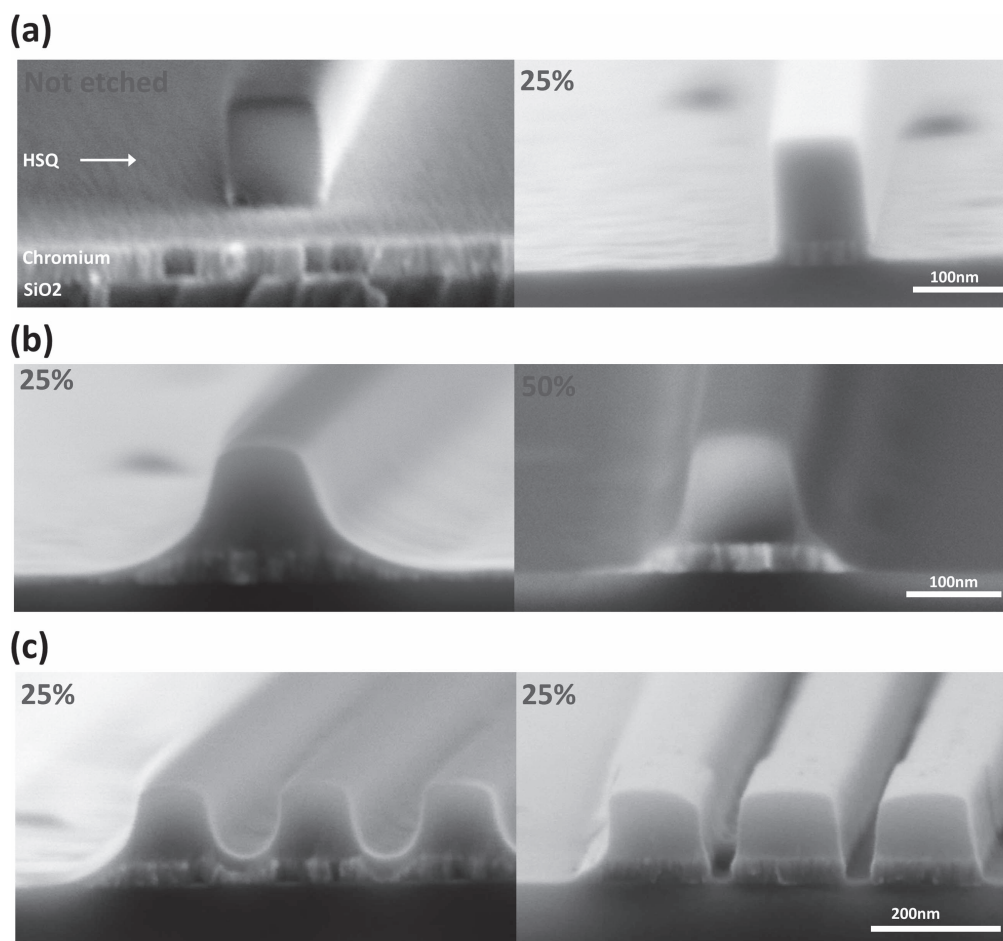


Figure 4. Patterned chromium etching at low temperatures. (a) Chromium before (left) and after (right) etching at -50°C in 25% oxygen gas mixture. (b) and (c) Etching at -100°C shows a deposit is formed close to the HSQ pattern. This slows down etching rates but also prevents etching of chromium from under the mask. (b) single lines etched at 25% and 50% oxygen. Note visible deposit on HSQ sidewall in right image. (c) Shows 200 nm pitch samples after 15 min (left) and 30 min (right) in the etch gas chemistry. (a) and (b) HSQ lines of 110 nm height and 100 nm width. (c) 35 nm trench width at the bottom.

in sticking coefficients with surface conditions. We believe the changes at the lowest temperatures shown, -80°C , is related to the volatility of the main etching product. The equilibrium vapor pressure of CrO_2Cl_2 , can be extrapolated from data by Stull's *et al* [40] using the Antoine correlation (supplement C). We find that for our ambient chamber pressure of 5 mTorr, the corresponding boiling point temperature is -74.8°C . This means that the chromyl chloride etching product is expected to be liquid below this point which leads to reduced etching rates.

Etching below -80°C with patterned features

Below -80°C etching rates were extremely slow. Work on patterned features showed that etching occurred but was complicated by deposition of a product visible at -90°C . Here we compare patterned features etched at -50°C and -100°C to highlight changes that occur while etching below the boiling point, near the freezing point of -96.5°C for chromyl chloride.

Figure 4 shows the etching mask and features etched at -50°C and -100°C . Etching at -50°C with 25% oxygen

for this type of feature and etch time (figure 4(a) right) produces a slight undercut and positive profile (our subsequent paper shows that this can be significantly improved). On the contrary, etching at -100°C leads to deposition of material, the amount of which is a function of the oxygen level. Figure 4(b) shows sidewall deposits on the HSQ and in the corners of the features. Figure 4(b) left is after 15 min of etching at 25% O_2 ratio while figure 4(b) right is after 15 min of etching at 50% oxygen. After chromium is gone in the open area, no deposit is visible on the SiO_2 interface far from the features. We assume the deposition materials only sticks well to itself and the HSQ mask, and therefore originates on the HSQ mask and then growth to cover the material in proximity to the features. Pure chromyl chloride would evaporate after returning to room temperature. We assume it is not pure chromyl chloride. Note that the freezing point of CrO_2Cl_2 is -96.5°C which is very close to the etch temperature. A freezing point depression could be present pushing it below -100°C . In figure 4(c) we show the process evolving in 200 nm pitch features. Figure 4(c) (left) chromium etching stopped after being completely passivated within the trench. Increasing etch time in figure 4(d) (right) breaks

through the passivant. The remaining chromium is etched away and leaves a very vertical profile. While etching rates are low, anisotropic profiles are evident, despite the known isotropic mechanism of chlorine/oxygen etching of chromium [41]. This patterning demonstration at -100°C further highlights how temperature is a strong knob to turn in the etching process, changing the mechanisms. Further work in this low temperature regime should be focused on an influence of the ion bombardment on the profile evolution. With this approach, we expect controlled removal of the deposited chromyl-chloride layer, and consequentially, control of the etching rates and anisotropy.

Note that in our etch gas chemistry the HSQ resist masks used here have advantages over traditionally used carbon based masks for chromium etching—with HSQ the selectivity to Cr is generally greater than 20 even at very high oxygen concentrations. In subsequent work we explore etching of chromium features as a function of temperature and oxygen content using HSQ as a mask. The high selectivity, with almost undetectable mask erosion, allows us to better characterize etching profile development.

4. Conclusions

We investigated chromium ICP etching under a wide range of Cl_2/O_2 gas mixture ratios and with changing electrode temperatures with the goal of understanding how to control etching for sub-5 nm patterning of thin films.

At the extremes of only Cl_2 or only O_2 gas flow, the etch rates were comparable to Argon plasma sputtering, far below 1 nm min^{-1} . Etching rates increased with the introduction of more oxygen into the Cl_2/O_2 mixture. Maximum rates were found between 20% and 30% of the oxygen partial pressure.

Changing temperature from $+40^{\circ}\text{C}$ to -80°C revealed a cross-over of etch rate curves in the low oxygen regime. That is, the rates at lower temperature were found to be higher than those at higher temperature. This is most likely due to an increase of the oxygen sticking coefficient promoting surface conditions more favorable to forming the chromyl chloride product.

Calculations of the total flux of species to the substrate in comparison with the number of etched chromium atoms under the assumption of only CrO_2Cl_2 product shows evidence that O^* and Cl^* radicals are likely to be the dominant reactants.

Arrhenius plots show very low activation energies (for oxidation of chromium for forming chromyl chloride), between 0.04 and 0.12 eV, with activating energy increasing with increasing oxygen concentration. Such low activation energies are in the range of physisorption processes. The Arrhenius plot at the mid oxygen levels (20%–50%), where the chromyl chloride is likely to form, show good linearity down to -50°C . At -80°C we are seeing a change most likely due to a phase change of the chromyl chloride.

Etching of HSQ masked chromium at -100°C shows redeposition of etch products with very high sticking probability to the HSQ mask forming a passivant on the chromium inside the pattern. This regime, may provide a means to

control anisotropy using ion bombardment to remove the deposition product directionally.

Overall, we find we have a method to control precisely change mechanisms and rates using both oxygen involvement and low temperatures as knobs.

Finally the next step is to study profile evolution using lithographic pattern for sub-5 nm control of chromium etching. In subsequent work we will show cryogenic etching gives advantages for controlling critical dimensions in etched chromium with control in the single nanometer regime. The blanket film studies lay an excellent foundation for identifying the best etching regime and for understanding etching mechanisms in patterned features. This puts us on the path for developing nanoimprint templates for bit patterned media at 5 Td/in^2 and beyond.

Acknowledgments

We are especially grateful to Seagate Technology LLC for support of DS. This work was completed at the Molecular Foundry and supported by the Office of Science, Office of Basic Energy Sciences, of the US Department of Energy under Contract No. DE-AC02-05CH11231. DO, SD, and SS, were supported by this contract. The work of D S was partially supported by the EU Contract No. [318804] [Single Nanometer Manufacturing for beyond CMOS devices (SNM)].

References

- [1] Curtis B J, Brunner H R and Ebnoether M 1983 Plasma processing of thin chromium films for photomasks *J. Electrochem. Soc.* **130** 2242–9
- [2] Flack W W, Tokunaga K E and Edwards K D 1993 Chrome dry-etching for photomask fabrication *Proc. SPIE* **1809** 85
- [3] Lin H C *et al* 2015 Deep ultraviolet laser direct write for patterning sol-gel InGaZnO semiconducting micro/nanowires and improving field-effect mobility *Sci. Rep.* **5** 1–11
- [4] Albrecht T R *et al* 2015 Bit-patterned magnetic recording: theory, media fabrication, and recording performance *IEEE Trans. Magn.* **51** 1–42
- [5] Larrieu G and Han X L 2013 Vertical nanowire array-based field effect transistors for ultimate scaling *Nanoscale* **5** 2437–41
- [6] Zhang Q *et al* 2004 Magnetization reversal of CrO_2 nanomagnet arrays *J. Appl. Phys.* **96** 7527–31
- [7] Felser C, Fecher G H and Balke B 2007 Spintronics: a challenge for materials science and solid-state chemistry *Angew. Chem.-Int. Ed.* **46** 668–99
- [8] Shiroishi Y *et al* 2009 Future options for HDD storage *IEEE Trans. Magn.* **45** 3816–22
- [9] Malloy M and Litt L C 2011 Technology review and assessment of nanoimprint lithography for semiconductor and patterned media manufacturing *J. Micro-Nanolithogr. MEMS MOEMS* **10** 032001
- [10] Wu B Q 2006 Photomask plasma etching: a review *J. Vac. Sci. Technol. B* **24** 1–15
- [11] Abe H *et al* 1976 Microfabrication of anti-reflective chromium mask by gas plasma *Japan. J. Appl. Phys.* **15** 25–31

- [12] Nakata H, Nishioka K and Abe H 1980 Plasma-etching characteristics of chromium film and its novel etching mode *J. Vac. Sci. Technol.* **17** 1351–7
- [13] Kwon K H *et al* 1999 Additive oxygen effects in Cl_2 plasma etching of chrome films *J. Mater. Sci. Lett.* **18** 1197–200
- [14] Dussart R *et al* 2014 Plasma cryogenic etching of silicon: from the early days to today's advanced technologies *J. Phys. D: Appl. Phys.* **47** 123001
- [15] Rangelow I W 2003 Critical tasks in high aspect ratio silicon dry etching for microelectromechanical systems *J. Vac. Sci. Technol. A* **21** 1550–62
- [16] Dussart R *et al* 2003 Passivation mechanisms in cryogenic SF_6/O_2 etching process *J. Micromech. Microeng.* **14** 190
- [17] Liu Z *et al* 2014 Low-temperature plasma etching of high aspect-ratio densely packed 15 to sub-10 nm silicon features derived from PS-PDMS block copolymer patterns *Nanotechnology* **25** 285301
- [18] Liu Z *et al* 2012 Super-selective cryogenic etching for sub-10 nm features *Nanotechnology* **24** 015305
- [19] Tachi S *et al* 1991 Low-temperature dry etching *J. Vac. Sci. Technol. A* **9** 796–803
- [20] Coleman T P, Buck P D and Johnson D J 1996 Plasma etching of Cr films in the fabrication of photomasks: II. A comparison of etch technologies and a first look at defects *Proc. SPIE* **2884** 92–102
- [21] Abe T, Yokoyama T, Kyoko S, Miyashita H and Hayashi N 1998 Comparison of etching methods for subquarter-micron-rule mask fabrication *Proc. SPIE* **3412** 163
- [22] Wu B 2003 Investigation of Cr etch kinetics *Proc. SPIE* **5256** 701
- [23] Kawada H *et al* 2001 *In situ* characterization of residues formed on a plasma-etching chamber *J. Vac. Sci. Technol. A* **19** 31–7
- [24] Nesladek P, Ruhl G G and Kristlib M 2004 Investigation of Cr etch chamber seasoning *European Mask and Lithography Conference (EMLC) 2004*
- [25] Dictus D *et al* 2010 Impact of metal etch residues on etch species density and uniformity *J. Vac. Sci. Technol. B* **28** 789–94
- [26] Hilfiker J N *et al* 2008 Survey of methods to characterize thin absorbing films with spectroscopic ellipsometry *Thin Solid Films* **516** 7979–89
- [27] Fujiwara H 2007 *Spectroscopic Ellipsometry: Principles and Applications* (Chichester, England; Hoboken, NJ: Wiley) 369p xviii
- [28] Huttner B 1995 On brewsters angle of metals *J. Appl. Phys.* **78** 4799–801
- [29] Kovalenko S and Lisitsa M 2001 Thickness dependences of optical constants for thin layers of some metals and semiconductors *Semicond. Phys. Quantum Electron. Optoelectron.* **4** 352–7
- [30] Kang S Y *et al* 2001 Etch characteristics of Cr by using Cl_2/O_2 gas mixtures with electron cyclotron resonance plasma *J. Electrochem. Soc.* **148** G237–40
- [31] Ichiki T, Takayanagi S and Horiike Y 2000 Precise chrome etching in downstream chlorine plasmas with electron depletion through negative ion production *J. Electrochem. Soc.* **147** 4289–93
- [32] Gaballah I, Ivanaj S and Kanari N 1998 Kinetics of chlorination and oxychlorination of chromium (III) oxide *Metall. Mater. Trans. A* **29** 1299–308
- [33] Schäfer H and Wartenpfehl F 1961 Das chrom (III)-oxydchlorid CrOCl *Z. Anorg. Allg. Chem.* **308** 282–9
- [34] Ebbinghaus B B 1995 Thermodynamics of gas-phase chromium species—the chromium chlorides, oxychlorides, fluorides, oxyfluorides, hydroxides, oxyhydroxides, mixed oxyfluorochlorohydroxides, and volatility calculations in waste incineration processes *Combust. Flame* **101** 311–38
- [35] Hoekstra R J and Kushner M J 1996 Predictions of ion energy distributions and radical fluxes in radio frequency biased inductively coupled plasma etching reactors *J. Appl. Phys.* **79** 2275–86
- [36] Fleddermann C and Hebner G 1998 Measurements of relative BCl density in BCl_3 -containing inductively coupled radio frequency plasmas *J. Appl. Phys.* **83** 4030–6
- [37] Tonoitani J, Ohmi S and Iwai H 2005 Dry etching of $\text{Cr}_2\text{O}_3/\text{Cr}$ stacked film during resist ashing by oxygen plasma *Japan. J. Appl. Phys.* **44** 114–7
- [38] Zhdanov V P 1989 The coverage dependence of the sticking coefficient and the desorption pre-exponential factor *Surf. Sci.* **209** 523–35
- [39] Vattuone L *et al* 1994 Coverage dependence of sticking coefficient of O_2 on $\text{Ag}(110)$ *J. Chem. Phys.* **101** 726–30
- [40] Stull D 1947 Inorganic compounds *Ind. Eng. Chem.* **39** 540–50
- [41] Buck P D and Grenon B J 1994 *Comparison of wet and dry chrome etching with the CORE-2564* *Proc. SPIE* **2087** 42